IBM Z DATATHON 2026 · TEAM DATABAES

Payment scams trick people into sending the money themselves.

Payment Shield: real-time payment scam prevention on IBM LinuxONE

£576.4m

lost to authorised push payment (APP) scams in the UK in 2025

248,070

APP scam cases in 2025, each one approved by the victim

17% → 28%

Scams that start with a call or text: 17% of cases, 28% of losses

Source: UK Finance, Annual Fraud Report 2026

WHO IT HURTS

The cost lands on people who can least absorb it

Older and isolated customers

Targeted by callers posing as the bank, the police or a family member.

People under financial stress

Lured by fake investments, loans and romance scams that promise a way out.

Small businesses

Lose whole invoices when a supplier’s bank details are quietly swapped.

Banks and communities

Carry reimbursement costs, call-centre load and lost trust in digital payments.

SOLUTION

Score every payment before the money moves

LOW RISK

Approve

The payment goes straight through and the customer sees “Payment approved”.

HIGH RISK

Pause and explain

The app stops and shows a risk score and why, such as a new payee while someone is on the phone.

CUSTOMER DECIDES

Confirm or cancel

The customer stays in control of their own money, with a moment to think.

Every score, reason and decision goes into an audit log, so analysts can see exactly what happened.

DEMO

Three payments, one audit trail

Payment	What the system sees	Outcome
£650 rent to the usual landlord	Known payee, usual amount, daytime	Payment approved
£2,400 to a new “safe account”	New payee, pasted details, on a call, nearly 4× usual	Paused with reasons, customer cancels
£600 to a long-standing payee	Six times the usual amount, while on a call	Paused, customer confirms

Then open the audit log · Backup: the recorded demo video

ARCHITECTURE

Scoring that sits inside the payment path

Payment in

Payment form posts to POST /assess on FastAPI, port 8000

Features

Amount vs usual, new payee and age, hour, device change, hesitation, pasted payee, on call

Rules + ML

Gradient Boosting model and red-flag rules, blended into a 0–100 score

Explanation

Top 3 reasons, ranked by feature importance, in plain English

Human confirm

Under 55 approved; 55 and over paused for Confirm or Cancel

Every score, reason and choice is written to a SQLite audit log, shown at /audit · all running on a LinuxONE VM

IBM Z USAGE

Why this belongs on LinuxONE

Built and run on LinuxONE

Model trained and the FastAPI app served on our LinuxONE VM (s390x), with Python, scikit-learn and SQLite.

Scoring inside the payment

Each model decision takes 0.46 ms (median) on our LinuxONE server, so scoring fits inside the payment.

Data stays on the platform

Training data, model and audit log never leave the LinuxONE VM.

Where payments already run

Many banks run core payments on IBM Z, so scoring next to the transaction data is the realistic path.

RESULTS · SYNTHETIC DATA

What the prototype achieved

97%

of scam payments caught (recall): 291 of 300

92.4%

of flagged payments were real scams (precision)

24

false alarms out of 9,700 genuine payments

0.61 ms

p99 time per model decision on LinuxONE

Rules alone caught **72.3%** of scams. The model catches **97%** on the same test set.

Synthetic data: 50,000 payments, 3% scams · 10,000 held-out test payments · scikit-learn gradient boosting · from results.json

IMPACT AND NEXT STEPS

Fewer victims, and a clear path to a pilot

Impact

- 97 in every 100 scam payments paused in our synthetic test
- Only 24 of 9,700 genuine test payments paused, so real customers rarely wait
- Warnings that teach customers to spot the next scam

Next steps

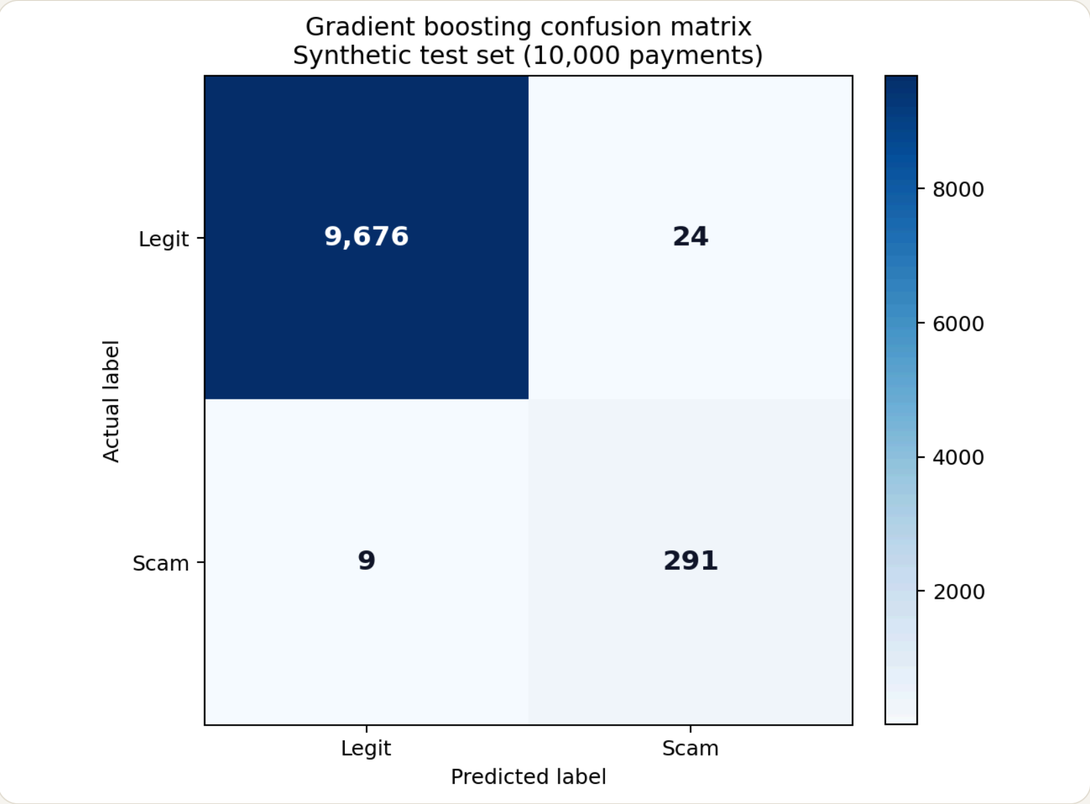
- Test on real, anonymised payment data with a bank or fintech partner
- Retrain on confirm and cancel choices from the audit log
- Run inference on IBM Z’s on-chip AI accelerator at production volume

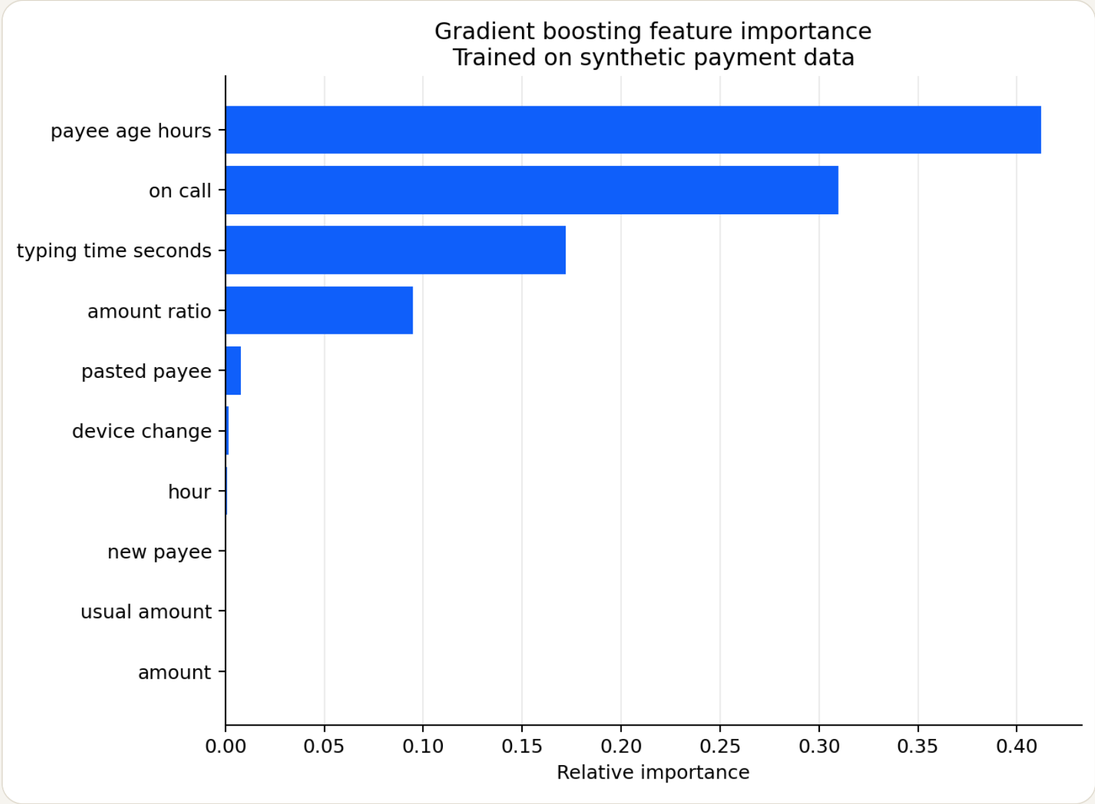
Thank you · Payment Shield by Team Databaes

github.com/36taransingh5-dotcom/ibm-z-datathon-2026-databaes · SSF2026

BACKUP · SYNTHETIC DATA

How the model decides





Both plots from generate_and_train.py on the 10,000-payment synthetic test set · evidence/ folder in our repo